

Yuxuan Ni

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EDUCATION

The University of Alabama Ph.D., Electrical and Computer Engineering Tuscaloosa, AL	Sep 2025 - Present
Carnegie Mellon University M.S. in Engineering & Technology Innovation Management Pittsburgh, PA	Jan 2023 - Dec 2023
UESTC / University of Glasgow B.E. in Communication Engineering; B.E. in Electronics and Electrical Engineering with Communications	Sep 2018 - Jun 2022

SELECTED RESEARCH PROJECTS

Bunker Multi-Sensor SLAM, Localization, and Mapping <i>Research Project Prof. Shunqiao Sun and Prof. Igal Bilik The University of Alabama / Ben-Gurion University</i> - Built a ROS2-based UGV perception and evaluation stack for bunker, low-light, and GNSS-degraded field runs, synchronizing camera, LiDAR, 4D radar, INS/RTK, wheel feedback, and driver-control data. - Developed real-time LiDAR localization and soft-anchor mapping workflows; reduced a 459.5M-point raw scan stream to a 2.13M-point map (99.5% reduction) and cleaned a 24.29M-point map to 13.56M points. - Improved radar localization through IMU-aided fusion and sensor-specific tuning, using INS/RTK alignment as the reference for trajectory evaluation and failure-case analysis. - Evaluated radar, visual, and LiDAR-inertial localization in a low-light, building-occluded sequence; radar odometry reached 7.16 m ATE RMSE versus 35.47 m for visual SLAM and 35.57 m for a LiDAR-inertial baseline.	Aug 2025 - Present
Learning-Based 4D Radar Odometry and Camera Fusion <i>Research Project Prof. Shunqiao Sun The University of Alabama</i> - Extended a learning-based 4D radar odometry pipeline for sparse radar point clouds, adding range/intensity pre-filtering and point-multiplication experiments to improve usable geometric structure before pose estimation. - Built two-stage odometry experiments that first stabilize radar pose outputs and then evaluate camera-pose fusion, including an MLP-based module for combining camera estimates with radar trajectories. - Added lightweight pseudo-correspondence supervision while preserving the original pose backbone; introduced 84.1K trainable parameters, about 0.53% of the full model, for correspondence and offset learning. - Reduced correspondence loss by 36.0%, match/static losses by about 36.8%, and offset error from 0.409 m to 0.363 m in frozen-backbone ablations; guarded refinement reduced translation RMSE from 0.752 m to 0.311 m.	Jan 2026 - Present
NXP Radar DDM Decoding <i>Collaborative radar signal-processing project</i> - Developed a MATLAB SFW/DDM simulation and decoding pipeline for 12-TX/12-RX, 1024-sample, 512-chirp radar data with multi-TX de-spreading and range-Doppler fusion. - Implemented CA-CFAR and DBSCAN-based target extraction to separate DDM ghost artifacts, SFW Doppler offsets, and velocity-induced Doppler smearing in staged validation scenarios. - Designed velocity-search and quadratic phase-correction desmearing, validating a 320-candidate search on close targets with 2 m range separation and 40 m/s velocity difference.	Aug 2025 - Present
MVMD-Based CSP for EEG Signal Classification <i>Research Assistant Prof. Aimin Jiang Hohai University</i> - Applied Multivariate Variational Mode Decomposition (MVMD) across 1,000+ EEG datasets to validate inter- and intra-subject signal variability. - Designed a Common Spatial Pattern (CSP) algorithm that improved classification accuracy by about 10%. - Optimized spatial and mode filters with ADMM and reduced computational complexity by about 20%. - Compared MLP, LDA, and SVM classifiers to validate feature robustness across EEG classification settings.	Jun 2024 - Feb 2025
NGSO Satellite In-line Interference Analysis <i>Research Assistant Prof. Jon Peha and Prof. Marvin Sirbu Carnegie Mellon University</i> - Simulated satellite-ground communication links with TLE data in MATLAB and Python, processing million-scale datasets and achieving location accuracy within 1 km. - Developed a custom simulation model to quantify in-line interference across satellite constellations under different communication protocols. - Enhanced simulation coverage for multiple satellite communication models, reducing computational cost by about 60%. - Designed and tested an interference-avoidance algorithm that reduced estimated interference probabilities by 30%.	Aug 2023 - Jan 2024
Low-Noise Balanced Photodetector for Single Photon Detection <i>Research Assistant Prof. Yunxiang Wang UESTC</i> - Reduced noise and enhanced sensitivity by integrating JFETs with photodiode circuits, targeting a 300 MHz theoretical bandwidth. - Selected and integrated a differential amplifier architecture for high-speed single-photon detection circuits. - Evaluated eight LTspice circuit designs and optimized PCB layouts in Altium Designer to reduce oscillation and improve circuit performance.	Sep 2021 - May 2022
5G Clustering Algorithm Simulation for Ultra-Dense Networks <i>Research Project Prof. Yuwei Wang Chinese Academy of Sciences</i> - Developed a clustering algorithm for 5G ultra-dense networks using path-loss mitigation and frequency-management strategies to reduce serial interference. - Validated the algorithm in NS3 simulations, achieving a reported 30% improvement in data reception rates.	May 2021 - Oct 2021

Design and Construction of a Multi-Function Rover

Feb 2021 - Jun 2021

Team Project | Communication Group Lead

- Led development of a low-latency wireless communication system for a multifunctional rover, supporting real-time data exchange with the control center.
- Implemented route tracking and color recognition functions with custom sensing and control logic, achieving over 90% navigation accuracy in project tests.
- Coordinated embedded programming, circuit design, and communication-module integration under project schedule constraints.

WORK EXPERIENCE

CMU Center for Technology Transfer and Enterprise Creation

May 2023 - Aug 2023

Enterprise Creation Intern | Pittsburgh, PA

- Built profiles for 1,000+ customers, identified 60+ high-potential leads, and connected with 26 key prospects for CMU Bio-robotic Lab market discovery.
- Analyzed one-million-plus data points with RStudio and Excel to evaluate market potential and align technical use cases with commercialization evidence.
- Led Phlux consulting work including 20+ customer interviews to refine core use cases and market-entry assumptions.
- Defined 10+ proprietary technology use cases to support product positioning and comparative market-feedback analysis.

Signify (China) Investment Co., Ltd.

Sep 2022 - Dec 2022

Hardware Engineer Intern | Shanghai, China

- Optimized power-circuit designs with active EMI filters and high-performance capacitors, reducing power consumption by 30%.
- Debugged and tested IoT lamp functionality, improving lighting accuracy and user response speed by refining control algorithms and response logic.

China Mobile Communication Group Co., Ltd.

Jul 2020 - Aug 2020

Engineering Maintenance Intern | Changzhou, China

- Supported planning and deployment of 10+ 5G base stations, expanding network coverage in three service areas.
- Resolved base-station issues involving signal interference, transmission delay, and energy efficiency, contributing to reduced network downtime.
- Assisted maintenance workflows for malfunctioning base stations and documented field-level troubleshooting outcomes.

PUBLICATION

Y. Ni and L. Shi, "Specific clustering scenarios and simple algorithm simulation in 5G environment," Proc. SPIE 12087, EIECT 2021, 120870M, 2021.

TECHNICAL SKILLS

Python, MATLAB, C/C++, ROS/ROS2, Linux, PCL, OpenCV, PyTorch, NumPy, STK, NS3, LTspice, Altium Designer.